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Mixer assembly removability arrangements for hand-held beverage dispensers

Abstract

An arrangement for selective removal of a mixer assembly from a hand-held beverage dispenser includes a handle body housing a piping system for dispensing post-mix beverages and a toe plate for dependently supporting the mixer assembly. Upwardly projecting mounting bosses have central orifices running end-to-end therethrough, and extending at the bottom end through the toe plate. Each orifice is axially aligned with a mounting hole of a mixer assembly as operably supported by the tow plate, and releasable fasteners provided top-to-bottom through the orifices engage the mounting holes to affix the mixer assembly to the toe plate.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to post-mix drink dispensing, and, more particularly, but not by way of limitation, to arrangements for hand-held beverage dispensers enabling ready removal or attachment of post-mix type drink dispenser assemblies from or to hand-held beverage dispensers.

2. Description of the Related Art

(2) Post-mix type hand-held beverage dispensers, such as are often referred to as bar guns or soda guns are well known conveniences in the food and bar service industries, enabling beverages to be prepared as they are dispensed by mixing together a beverage product, such as, for example, a syrup or like concentrate, and a diluent, such as, for example, plain water or carbonated water. Although prior art dispensers of the bar gun class include such conveniences as readily removable nozzle housings, as are generally cooperatively adapted with a mixer assembly in implementation of a post-mix type drink dispenser assembly, conventional dispensers of this class generally lack features for maintainability of the remainder of the provided hand-held beverage dispenser. For

example, the mixer assembly, as distinguished from the cooperatively adapted nozzle housing, are generally difficult to adequately maintain and not readily removable for repair or replacement, let alone for routine cleaning. The implemented mixer assemblies are generally either unitary with the handle body or so highly integral therewith as to make removal or disassembly for maintenance impractical if not wholly impossible.

(3) Accordingly, a hand-held post-mix type drink dispenser including a readily removable mixer assembly will facilitate repair or replacement of the mixer assembly, and will promote good health by enabling thorough cleaning on a routine basis, and will enable flexibility in the development of a commercial offering by allowing use of any number of differently designed dispenser assemblies with a single design of handle body and piping system.

SUMMARY OF THE INVENTION

(4) An arrangement for selective removal of a post-mix type mixer assembly from a hand-held beverage dispenser includes a handle body housing a piping system for dispensing post-mix beverages through a mixer assembly; a toe plate for dependently supporting the mixer assembly; and a plurality of upwardly projecting mounting bosses formed on a top side of the toe plate, each mounting boss having a central orifice running end-to-end therethrough and extending from the bottom end of the mounting boss through the toe plate. Each orifice is adapted to be axially aligned with a mounting hole of a mixer assembly as operably supported by the toe plate. Releasable fasteners provided top-to-bottom through the orifices engage the mounting holes of the mixer assembly to affix the mixer assembly to the toe plate.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a top isometric view illustrating a mixer assembly removability arrangement according to the first embodiment.

(2) FIG. 2 is a side elevational view of the mixer assembly removability arrangement according to the first embodiment.

(3) FIG. 3 is a partially exploded top isometric view illustrating the mixer assembly removability arrangement according to the first embodiment.

(4) FIG. 4 is a partially exploded side elevational view illustrating the mixer assembly removability arrangement according to the first embodiment.

(5) FIG. 5 is a partial cross-sectional isometric view taken along lines 5-5 of FIG. 4.

(6) FIG. 6 is a partially exploded top isometric view illustrating a mixer assembly removability arrangement according to the second embodiment.

(7) FIG. 7 is a partially exploded bottom isometric view illustrating the mixer assembly removability arrangement according to the second embodiment.

(8) FIG. 8 is a partially exploded bottom isometric view illustrating various details of the mixer assembly removability arrangement according to the second embodiment.

(9) FIG. 9 is a partially exploded right side elevational view illustrating various details of the mixer assembly removability arrangement according to the second embodiment.

(10) FIG. 10 is a partially cut away bottom front end view taken along lines 140-140 of FIG. 139 illustrating various details of the mixer assembly removability arrangement according to the second embodiment.

(11) FIG. 11 is a top plan view illustrating various details of an exemplary mixer assembly suitable for use in the exemplary implementation of the mixer assembly removability arrangement according to the second embodiment.

(12) FIG. 12 is a rear end elevational view illustrating a cover of the mixer assembly removability arrangement according to the second embodiment.

(13) FIG. 13 is a partially cut away top front end view taken along lines 143-143 of FIG. 139 illustrating various details of the mixer assembly removability arrangement according to the second embodiment.

(14) FIG. 14 is a partially exploded bottom isometric view illustrating a mixer assembly removability arrangement according to the third embodiment.

(15) FIG. 15 is a top isometric view illustrating the mixer assembly removability arrangement according to the first embodiment.

(16) FIG. 16 is a bottom isometric view illustrating the mixer assembly removability arrangement according to the first embodiment.

(17) FIG. 17 is a partially exploded bottom isometric view illustrating the mixer assembly removability arrangement according to the first embodiment.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(18) As required, detailed embodiments of the present invention are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary of the invention, which may be embodied in various forms. Figures are not necessarily to scale, and some features may be exaggerated to show details of particular components or steps.

(19) As shown in FIGS. 1-5, a first embodiment of a mixer assembly removability arrangement is particularly suited to implementation in connection with a hand-held beverage dispenser 150 having a full length upper enclosure member 300 about the implemented handle body 151. As shown in FIGS. 1-2, such a suitable hand-held beverage dispenser 150 is shown with a mixer assembly 105 secured operably in place. In order to access the mixer assembly removability arrangement of the first embodiment, the full length upper enclosure member 300 about the implemented must first be removed to expose the forward end 156 of the handle body 151. As shown in FIG. 3, the full length upper enclosure member 300 about the handle body 151 of the exemplary hand-held beverage dispenser 150 is readily removed upon conventional extraction of the two upper enclosure mounting screws 307. With the upper enclosure member 300 lifted off the handle body 151, access ports 320 provided in accordance with the first embodiment are exposed at the forward end 156 of the handle body 151.

(20) As shown in the partially exploded view of FIG. 4, provided mixer mounting hardware such as, for example, machine screws 315 are readily conventionally extracted through the exposed access ports 320, thereby readily releasing the mixer assembly 105 from the handle body 151. As shown in the partial cross sectional view of FIG. 5, mixer assembly removability arrangement of the first embodiment includes the provision of preferably two upwardly projecting mounting bosses 316, each of which has a central through hole 319 extending beyond the mounting boss 316 and through the toe plate 364. In a critical aspect of the present invention, each of the through holes 319 is axially aligned with attached position of a respective threaded hole 130 at the top 107 of the mixer assembly 105. Additionally, however, collar 318 is most preferably provided about the top end 317 of each upwardly projecting mounting boss 316. The each provided collar 318 forms a rigid stop for the head of a machine screw 315 or like structure of alternatively implemented mixer mounting hardware, which in turn establishes a precisely known, fixed insertion depth for the machine screws 315 or other mounting hardware 315, and thereby preventing damage to the mixer assembly 105.

(21) Where a full length upper enclosure for a hand-held beverage dispenser is not possible or otherwise undesired, thereby precluding implementation of the first embodiment, a second embodiment of the mixer assembly removability arrangement contemplates access to interior space at the top, forward end of the handle body of a the hand-held beverage dispenser through an inventive cover. Turning now then to FIGS. 6-13, a structural arrangement for simple removal or attachment of a post-mix type drink dispenser assembly from or to a hand-held beverage dispenser—a mixer assembly removability arrangement—is shown and described. For convenience and clarity the exemplary implementation of the mixer assembly removability arrangement, as now set

forth, shows the mixer assembly removability arrangement as integrated with, and describes the arrangement with reference to, a hand-held beverage dispenser **150** that is specially adapted to implement the inventive mixer assembly removability arrangement, but is otherwise an exemplary only generic representative of the bar gun class of dispensers. That said, the inventive aspects of the mixer assembly removability arrangement are in no manner necessarily limited to application in connection with the now described hand-held beverage dispenser **150**, the exemplary post-mix type drink dispenser assembly, or any other particular implementation of a hand-held beverage dispenser or post-mix type drink dispenser assembly.

(22) On the other hand, the available space within, as determined by, the extents of the handle body **151** is necessarily very limited by such matters of commercial suitability as include the need to produce the handle body **151** of the hand-held beverage dispenser **150** with an ergonomically acceptable shape within the maximum dimensions acceptable to meet at least minimum requirements for ease of use and comfort in hand of the hand-held beverage dispenser **150**. As also previously discussed, the extents of the handle body **151** of a commercially suitable hand-held beverage dispenser **150** are further limited by the need to produce a hand-held beverage dispenser **150** having a total mass at or below the maximum total mass deemed acceptable for ease of use and comfort in hand of the hand-held beverage dispenser **150**. With this ever persistent matter of available space afforded by the extents of an otherwise optimized handle body **151** clearly in mind, however, it will be appreciated that Applicant has, within all of the foregoing constraints, devised a mixer assembly removability arrangement, as now described, that not only achieves very simple removal and attachment of a post-mix type drink dispenser from or to a hand-held beverage dispenser, but also advances the basic capabilities expected of hand-held beverage dispensers.

(23) As will be better understood further herein, Applicant's most preferred implementation of the mixer assembly removability arrangement goes beyond enabling simple removal or attachment of a single specially adapted post-mix type drink dispenser assembly from or to a hand-held beverage dispenser to use this inventively provided capability to further provide for use of any user-selected one of a plurality of post-mix type drink dispenser assemblies in connection with a single implementation of a hand-held beverage dispenser. The plurality of post-mix type drink dispenser assemblies from which the user selection is made may include post-mix type drink dispenser assemblies of differing operating characteristics, such as may be desirable for primarily dispensing one type of beverage product as opposed to another. Additionally and as will also be better understood further herein, the plurality of post-mix type drink dispenser assemblies from which the user selection is made may also include commercially available off-the-shelf post-mix type drink dispenser assemblies, including assemblies originally developed for use with countertop or standalone post-mix drink dispensers. What is more, compatibility features incorporated within the implemented mixer assembly removability arrangement, as now shown and described or that may be later developed, may enable use of such commercially available off-the-shelf post-mix type drink dispenser assemblies with little or no modification or other adaptation, while nonetheless meeting the previously described design constraints.

(24) Referring now to FIGS. **6-7**, in particular, each of which depicts various components of the hand-held beverage dispenser **150** as exploded along the longitudinal axis **140** through the mixer assembly **105**, an exemplary mixer assembly **105** for use with the hand-held beverage dispenser **150** is shown as positioned for operable dependent attachment to the bottom **154** of the handle body **151** of the hand-held beverage dispenser **150** at a location adjacent the forward end **156** of the handle body **151**. Additionally, the figures show a specially formed cover **352** for an opening **351** to an interior space **350** formed or otherwise provided near the top **152** and at the forward end **156** of the handle body **151** of the hand-held beverage dispenser **150**. The specially formed cover **352** is shown as positioned for placement in use to cover the opening **351** to the interior space **350**. As also shown in FIG. **6**, but more clearly in FIG. **13**, the opening **351** provides access to a plurality of upwardly projecting mounting bosses **366**, which are also specially formed within the interior

space **350**. As will be better understood further herein, the cover **352** for the opening **351** to the interior space **350**, positioned as described with respect to the opening **351**, is also simultaneously positioned for cooperative engagement with the upwardly projecting mounting bosses **366** in implementation of the inventive mixer assembly removability arrangement of the present invention. (25) FIGS. **6-7** also show various other components of the hand-held beverage dispenser **150**, as support implementation of the mixer assembly removability arrangement, or form part thereof. For example, a plurality of connector tubes **134** is shown as operably deployed within the various previously described fluid outlets **174** from the piping system of the hand-held beverage dispenser **150**. As will be additionally described further herein, the connector tubes **134** facilitate the previously mentioned attachment of the mixer assembly **105** to the handle body **151**. A rubber or like material seal **133** is shown as positioned for use in operative attachment of the mixer assembly **105** to the handle body **151**, as also will be better understood further herein. Finally, the figures show a plurality of machine screws **363** as positioned for use in operably affixing the specially formed cover **352** in place about the opening **351** to the previously discussed interior space **350**. Implementing multi-functionality similar to that as previously noted for the cover **352** and as will be better understood further herein, the machine screws **363**, or other suitably provided fasteners, positioned as described with respect to the cover **352**, are also simultaneously positioned for cooperative engagement with the previously discussed specially formed cover **352**, upwardly projecting mounting bosses **366**, elements of the mixer assembly **105** and additional elements of the handle body **151**, as will be described further herein, in implementation of the inventive mixer assembly removability arrangement of the present invention.

(26) In order to doubly ensure clarity in the following further description of the inventive mixer assembly removability arrangement of the present invention, it is noted that the components and the operation of the exemplary mixer assembly **105** are shown and described in greater detail in U.S. patent application Ser. No. 16/394,889 filed Apr. 25, 2019 for METHODS AND APPARATUS FOR POST-MIX DRINK DISPENSING, which by this reference is incorporated herein as though herein now set forth in its entirety.

(27) As shown in FIG. **6**, a plurality of connector tube orifices **108**, as will be detailed further herein, is provided at the top **107** of the mixer assembly **105** for introduction of beverage components to the mixer assembly **105**. As will be better understood further herein, each beverage product conveyed through one of the beverage product dispensing conduits **173**, as shown for example in FIG. **13**, is ultimately introduced to the mixer assembly **105** through a corresponding one of the connector tube orifices **108** at the top **107** of the mixer assembly **105**, wherein the introduced beverage product is channeled internally through the mixer assembly **105** as is otherwise conventional in the relevant arts. To the extent that additional provision must be made to convey fluids from the fluid outlets **174** to the connector tube orifices **108** at the top **107** of the mixer assembly **105**, however, a plurality of connector tubes **134** as shown in FIGS. **7-9** are provided, as will be better understood further herein.

(28) As shown in FIG. **10**, the preferred implementation of the hand-held beverage dispenser **150** comprises a toe plate **364**, which, as will be better understood further herein, provides a sturdy base upon which the interface between the piping system of the exemplary hand-held beverage dispenser **150** and the mixer assembly **105** is implemented. The toe plate **364** is preferably unitarily formed with the handle body **151**, and in any case at the bottom of the interior space **350** at the forward end **156** of the handle body **151**. As shown in FIG. **8**, the toe plate **364** is supported all around by, and forms a part of, the bottom **154** of the handle body **151**.

(29) In any case and as best shown in FIG. **13**, the terminus of each beverage product dispensing conduit **173** is collected, arranged and stably fixed to the top side **365** and slightly within the toe plate **364**. As will in light of this exemplary description be appreciated by those of ordinary skill in the relevant arts, the collective support provided by the rigidly provided toe plate **364** prevents any excessive strain or stress on or to the beverage product dispensing conduits **173** as the mixer

assembly **105** of the post-mix type drink dispenser assembly is attached or removed to or from the handle body **151**. Additionally and as best shown in FIG. **10**, each of the fluid outlets **174** from the piping system is formed within the toe plate **364** at the terminus of a corresponding one of the beverage product dispensing conduits **173**. As shown in FIGS. **7-8** and each of the provided fluid outlets **174** extends downward through the toe plate **364**, and opens to the exterior of the handle body **151** at the bottom side **371** of the toe plate **364** through the substantially planar top of a downwardly opening cylindrical cavity **141** formed in the bottom side **371** of the toe plate **364**. As will in light of this exemplary description also be appreciated by those of ordinary skill in the relevant arts, fixing the fluid outlets **174** about the depicted substantially planar top of the downwardly opening cylindrical cavity **141** of the toe plate **364** provides a consistent rigid interface for attachment of the mixer assembly **105** of the post-mix type drink dispenser assembly.

(30) In any case, the fluid outlets **174** from the piping system **158** are arranged at the bottom side **371** of the toe plate **364** as required to interface with the mixer assembly **105** of the post-mix type drink dispenser assembly. Additionally, the previously noted arrangement at the top side **365** of the toe plate **364** of the terminus of each beverage product dispensing conduit **173** most preferably corresponds to the arrangement of the fluid outlets **174** at the bottom side **371** of the toe plate **364**. In this manner, implementing the previously discussed fluid communication through the toe plate **364** between the beverage product dispensing conduits **173** at the top side **365** and the corresponding fluid outlets **174** at the bottom side **371** is facilitated.

(31) As will be better understood herein, the previously described cylindrical cavity **141** is sized and otherwise adapted to dependently receive the upper portion of the mixer assembly **105**, as operably affixed to the handle body **151** of the hand-held beverage dispenser **150**. As shown in the figures and particularly in FIG. **10**, each fluid outlet **174** comprises a cylindrical orifice formed or otherwise provided in the top of the downwardly opening cylindrical cavity **141** of the toe plate **364** concentrically about the terminus of a corresponding one of the product dispensing conduits **173**. The fluid outlets **174**, in turn, are arranged as required to interface with the mixer assembly **105**, and the cylindrical orifices forming the fluid outlets **174** from the piping system **158** are also sized, shaped and otherwise adapted as required to interface with the mixer assembly **105** of the post-mix type drink dispenser assembly, as will be better understood further herein.

(32) Providing for efficient yet reliable attachment of any post-mix type drink dispenser assembly to a hand-held beverage dispenser can be difficult under any circumstance. Providing a hand-held beverage dispenser with an efficient and reliable attachment interface conforming to an off-the-shelf post-mix type drink dispenser assembly is exceptionally challenging. In order to take advantage of manufacturer experience in achieving leak and other trouble free attachment of a post-mix type drink dispenser assembly, or a product line of post-mix type drink dispenser assemblies, with minimal, if any, modification of the post-mix type drink dispenser assembly and ever cognizant of the need to adhere to the previously detailed general constraints for implementation of a commercially suitable hand-held beverage dispenser **150**, which address such matters as size, mass, comfort in hand, ease of use and maintenance, safety and reliability, Applicant has determined that securement of the post-mix type drink dispenser assembly **104** from the top **152** of the handle body **151** is most desirable.

(33) As particularly shown in FIG. **13**, however, the interior space **350** at the forward end **156** of the handle body **151**, at the bottom of which the toe plate **364** is located, is like all other spaces substantially within the extents of the handle body **151** nearly fully occupied by elements of the piping system for the hand-held beverage dispenser **150**. As shown in the figures, each of the beverage product dispensing conduits **173** run horizontally into the interior space **350**, and then vertically downward to interface as previously described with the toe plate **364**. As will be appreciated by those of ordinary skill in the relevant arts, the respective points of entry to the interior space **350** of each of the beverage product dispensing conduits **173** is largely predetermined by the needs to accommodate valves, valve controls and the like features within the limited extents

of the handle body **151**. Additionally, the respective points of interface with the toe plate **364** of each of the beverage product dispensing conduits **173**, as well as the size and to some extent shape of the toe plate **364**, are largely determined by arrangement about the top of a compatible mixer assembly **105** of beverage component inlets, as well as the size and shape of the mixer assembly **105**.

(34) As a result of the foregoing parameters, the ability to modify the implemented routing through the interior space **350** of each of the beverage product dispensing conduits **173** without expanding the extents of the handle body **151** is very limited at best. Correspondingly, the ability to combine or otherwise rearrange unused area at the top side **365** of the toe plate **364** is likewise very limited at best. As particularly shown in FIG. **11**, the top **107** of the mixer assembly **105** of the implemented post-mix type drink dispenser assembly comprises a plurality of tapped holes **130** for dependently affixing the mixer assembly **105** to a beverage dispenser, each of which is longitudinally oriented parallel to the longitudinal axis **140** through the mixer assembly **105**. Utilizing these manufacturer provided features, in accordance with the most preferred implementation of the mixer assembly removability arrangement of the present invention, machine screws of the correct thread size must be inserted, in alignment with the longitudinal axis of each tapped hole **130**, from within the interior space **350**, through the toe plate **364** and into each of the tapped holes **130** to a specified insertion depth, notwithstanding the above described limited unused area at the top side **365** of the toe plate **364**, and the general difficulty of physically reaching any unused area at the top side **365** of the toe plate **364** as barricaded by the previously described routing of the beverage product dispensing conduits **173**.

(35) In addition to the foregoing obstacles to securing the mixer assembly **105** from the top **152** of the handle body **151**, it is noted that securement from the top **152** not only requires an opening **351** to the interior space **350**, but also provision of the previously mentioned cover **352** for the opening **351**. Additionally, the cover **352** must be readily removable, yet secure in use, thereby presenting the further obstacle that attachment hardware must be provided as well as appropriate structures within the interior space **350** for engagement of the hardware. Furthermore, the cover **352** must be accommodated at the depicted steep slope in order to avoid adding to the extents of the handle body **151**. Locating the cover **352**, for example at the forward end **156** or along a side **155** of the handle body **151** are also impracticable, as such access would fail to provide access for a screwdriver or like tool, along the longitudinal axis of each previously described tapped hole **130** for the mixer assembly **105**. Notwithstanding each of these obstacles to securement of the post-mix type drink dispenser assembly **104** from the top **152** of the handle body **151**, Applicant has implemented a novel and inventive mixer assembly removability arrangement enabling the desired functionality.

(36) As best shown in FIGS. **6-7** and **12-13**, a preferred implementation of the mixer assembly removability arrangement of the present invention generally comprises a specially formed cover **352** for an opening **351** to the previously described interior space **350** of the handle body **151** of the hand-held beverage dispenser **150**. A plurality of specially formed upwardly projecting mounting bosses **366** are disposed, within the interior space **350**, about the top side **365** of the toe plate **364**, which in substantial part forms the bottom of the interior space **350**, as particularly shown in FIG. **13**. As particularly shown in FIG. **10**, a plurality of clearance holes **369** is formed through the toe plate **364**, and aligned and arranged about the toe plate **364** complementary to the arrangement of the previously described tapped holes **130** about the top **107** of the mixer assembly **105** of the implemented post-mix type drink dispenser assembly. Each of the upwardly projecting mounting bosses **366** disposed about the top side **365** of the toe plate **364** is positioned about the top side **365** of the toe plate **364** in longitudinal alignment with one of the clearance holes **369** through the toe plate **364**.

(37) As shown in FIG. **13**, a counterbore **368** is formed or otherwise provided in the top end **367** of each of the preferably cylindrical upwardly projecting mounting bosses **366**. As also shown in the

figure, a corresponding one of the previously described clearance holes **369** through the toe plate **364** extends through each upwardly projecting mounting boss **366** to form a shoulder **370** at the bottom of the counterbore **368**. As shown FIGS. **10** and **13**, the shoulder **370** defines the first end of a clearance hole **369**, which runs through the bottom of the upwardly projecting mounting boss **366** at the top side **365** of the toe plate **364**, and continues through the toe plate **364** where the bottom end **139** of the clearance hole **369** emerges at the bottom side **371** of the toe plate **364**.

(38) As shown in FIGS. **6-7** and **12**, each previously mentioned specially formed cover **352** comprises a top, exterior face **353** and a bottom, interior face **355**. As particularly shown in FIG. **6**, a plurality of cylindrical hardware recesses **354** is formed through the top, exterior face **353** of the cover **352**. Importantly, each of the cylindrical hardware recesses **354** is longitudinally oriented parallel to the longitudinal axis **140** through the mixer assembly **105**. As shown in FIGS. **7** and **12**, a downwardly projecting mounting boss **356** is formed or otherwise provided from the bottom, interior face **355** of the cover, each downwardly projecting mounting boss **356** being longitudinally aligned with a corresponding one of the hardware recesses **354**. As shown in FIG. **12**, each of the hardware recesses **354** comprises a counterbore **357** having an interior diameter slightly larger than the head diameter of each of the machine screws **363** specified for the mixer assembly removability arrangement, as will be better understood further herein. A centrally aligned clearance hole **358** originates at the bottom of the counterbore **357**, thereby forming an interior shoulder **359** at the bottom of the counterbore **357**, and runs through the lower extent of the downwardly projecting mounting boss **356**. An exterior shoulder **361** is formed or otherwise provided at the bottom end **360** of the downwardly projecting mounting boss **356**, thereby forming a tubular insert **362**.

(39) Finally, the clearance hole **358** running through each downwardly projecting mounting boss **356** should be of equal diameter to that of the clearance hole **369** running through each upwardly projecting mounting boss **366**, and which diameter should be slightly larger than the diameter of the major threads of the machine screws **363** specified for the mixer assembly removability arrangement. Additionally, the outer diameter of each tubular insert **362** at the bottom end **360** of each downwardly projecting mounting boss **356** should be slightly less than the diameter of the counterbore **357** into each upwardly projecting mounting boss **366**, and the length of the tubular insert **362** at the bottom end **360** of each downwardly projecting mounting boss **356** should be slightly less than the depth of the counterbore **357** into each upwardly projecting mounting boss **366**. Finally, the outer diameter of each downwardly projecting mounting boss **356** should be coextensive with the outer diameter of each upwardly projecting mounting boss **366**.

(40) In use of the mixer assembly removability arrangement for firmly securing in place a mixer assembly **105**, as partially inserted into the cavity **141** and pressed snugly against the rubber seal **133**, the specially formed cover **352** is set in place over and about the opening **351** to the interior space **350**. As the cover **352** is set in place, it is noted that the tubular insert **362** at the bottom end **360** of each downwardly projecting mounting boss **356** will be received within the counterbore **357** into each corresponding upwardly projecting mounting boss **366**.

(41) At this juncture, it is noted that the lengths of each downwardly projecting mounting boss **356** and each corresponding upwardly projecting mounting boss **366** are most preferably implemented such that each tubular insert **362** is fully engaged within the corresponding counterbore **357**, and, of greater importance, each interior shoulder **359** formed in the downwardly projecting mounting bosses **356** lies in a single plane P31 transverse to the longitudinal axis **140** through the mixer assembly **105**, as attached to the handle body **151**. As will in light of this exemplary description be understood by those of ordinary skill in the relevant arts, the described arrangement enables specification of a single length for each machine screw **363** for use in the mixer assembly removability arrangement.

(42) In any case, the machine screws **363** are then inserted through the hardware recesses **354** at the top, exterior face **353** of the cover **352**. Each set of end-to-end combined clearance holes **358**, **369** provides a channel completely through the handle body **151**, the bottom of which is aligned

adjacent with and directly over a corresponding one of the tapped holes **130** about the top **107** of the first stage body **106** of the mixer assembly **105**. With the mixer assembly **105** manually supported as necessary, the machine screws **363** are tightened as specified for the mixer assembly removability arrangement, compressing the rubber seal **133** operatively between the top **107** of the mixer assembly **105** and the bottom side **371** of the toe plate **364**.

(43) As will be appreciated by those of ordinary skill in the relevant arts, the inventive mixer assembly removability arrangement provides for simple removal or attachment of a post-mix type drink dispenser assembly from or to a hand-held beverage dispenser. Additionally, the inventive mixer assembly removability arrangement enables the use of commercially available off-the-shelf and similar integrations in implementing a post-mix type drink dispenser assembly **104** for the hand-held beverage dispenser **150**. Still further, the inventive mixer assembly removability arrangement meets each object of the invention without any need for attachment collars or other interfaces, which would add very undesirable weight and bulkiness to the hand-held beverage dispenser **150**. The inventive mixer assembly removability arrangement also avoids the introduction of mounting hardware and the like to any part of the hand-held beverage dispenser **150** that in use is placed over and about a beverage vessel receiving a post-mix beverage, thereby eliminating the possibility of such hardware loosening and falling into a beverage vessel.

(44) Where a full length upper enclosure for a hand-held beverage dispenser is not possible or otherwise undesired, thereby precluding implementation of the first embodiment, and the interior space at the top, forward end of the handle body of a the hand-held beverage dispenser cannot provide sufficient space or clearance to implement the teachings of the second embodiment or provision of an access opening to the interior space is either not possible or otherwise undesired, a third embodiment of a mixer assembly removability arrangement contemplates provision of a mounting body for attachment of the mixer assembly **105** at the bottom of a hand-held beverage dispenser **10**.

(45) In accordance with the third embodiment, a mixer assembly removability arrangement for a hand-held beverage dispenser **10** comprises a preferably aesthetic mounting body **118** formed unitary or integral with a mixer assembly **105**, and including a provision for ready removal or attachment of the mounting body **118** from or to a hand-held beverage dispenser. As shown in FIGS. **14-16**, the mounting body **118** and at least a portion of the bottom adjacent the front end of the handle body and the aesthetic mounting body **118** are sized, shaped and otherwise cooperatively adapted for mating engagement one with the other. To this end, as shown in FIGS. **14-16**, mounting through holes **119** are provided through the aesthetic mounting body **118** corresponding to mating threaded holes **19** provided on the handle body **11**. As particularly shown in FIG. **17**, conventional mounting hardware **120** such as, for example, screws **121** pass through the mounting holes **119** disposed about the mounting body **118**, and securely engage the corresponding threaded holes **19** on the handle body **11**. As will be readily appreciated by those of ordinary skill in the art, the readily affixed mounting body **118**, including the provided mixer assembly **105**, is likewise readily removable from the handle body **11**.

(46) In order to facilitate attachment of the mixer assembly **105** in fluid communication with the plurality fluid outlets **36** of the handle body **11**, as particularly shown in FIGS. **14** and **17**, Applicant has found it desirable to provide an equal plurality of connector tubes **101** as are provided fluid outlets **36**. In accordance with the exemplary preferred implementation of the mixer assembly removability arrangement for the hand-held beverage dispenser **10**, a connector tube **101** is securely received in each of the fluid outlets **36** from the handle body **11**, such as by press fitting or the like. While the provided connector tubes **101** are preferably securely attached to the fluid outlets **36**, they are most preferably readily removably, but sealingly, received within corresponding connector tube orifices **108** provided in the implemented mixer assembly **105**.

(47) As previously noted, provision of the mixer assembly removability arrangement of the third embodiment turns squarely away from the longstanding practice in the prior art of providing a

mixer assembly that is either unitary with the handle body or so highly integral therewith as to make removal or disassembly for maintenance impractical if not wholly impossible. In at least the most preferred implementations of the third embodiment of the mixer assembly removability arrangement, however, any readily separable components of the mixer assembly **105**, such as, for example, multiple stage bodies, are press fit together or otherwise made integral. In this manner, service industry workers, as will typically care for the hand-held beverage dispenser **10**, will not be burdened with multiple parts that are easily lost or troublesome to assemble, while nonetheless making available the benefits of a removable mixer assembly **105**. As previously noted the details of an exemplary post-mix type drink dispenser assembly, including a mixer assembly which operates in a manner consistent with the present invention, are shown and described in U.S. patent application Ser. No. 16/394,889 filed Apr. 25, 2019 for METHODS AND APPARATUS FOR POST-MIX DRINK DISPENSING, which by this reference is incorporated herein as though now set forth in its entirety.

Claims

1. An arrangement for selective removal of a post-mix type mixer assembly from a hand-held beverage dispenser, said arrangement comprising: a handle body housing a piping system configured to dispense post-mix beverage products through a mixer assembly; a toe plate formed within said handle body, said toe plate being configured to dependently support said mixer assembly; a plurality of mounting bosses, each said mounting boss projecting upwardly from said toe plate and having a central orifice; wherein: each central orifice through a mounting boss extends from a bottom end of said mounting boss through said toe plate; and each central orifice is axially aligned with a mounting hole formed in said mixer assembly as operably supported by said toe plate; said handle body including a plurality of access ports, each said access port being axially aligned with one of each said central orifice through a mounting boss; and each said access port allowing passage for a releasable fastener provided top to bottom through each said central orifice, each said releasable fastener being adapted to releasably engage a said mounting hole formed in said mixer assembly.
2. The arrangement for selective removal of a post-mix type mixer assembly from a hand-held beverage dispenser of claim 1, comprising an upper enclosure member securable with the handle body thereby covering said plurality of access ports.
3. An arrangement for selective removal of a post-mix type mixer assembly from a hand-held beverage dispenser, said arrangement comprising: a handle body housing a piping system configured to dispense post-mix beverage products through a mixer assembly; a toe plate formed within said handle body, said toe plate being configured to dependently support said mixer assembly; a plurality of mounting bosses, each said mounting boss projecting upwardly from said toe plate and having a central orifice; wherein: each central orifice through a mounting boss extends from a bottom end of said mounting boss through said toe plate; and each central orifice is axially aligned with a mounting hole formed in said mixer assembly as operably supported by said toe plate; said handle body including an opening into an interior space having said plurality of upwardly projecting mounting bosses therein; a cover including a plurality of mounting bosses, each said mounting boss projecting downwardly from said cover and having a central orifice, said cover being configured to set in place over and about said opening to said interior space wherein each central orifice through a downwardly projecting mounting boss axially aligns with one of each said central orifice through an upwardly projecting mounting boss; and wherein a releasable fastener is provided top to bottom through each said central orifice of a downwardly projecting mounting boss and each said central orifice through an upwardly projecting mounting boss, each

said releasable fastener being adapted to releasably engage a said mounting hole formed in a mixer assembly.
